

Nurse Shift Scheduling System



Deepikaa Chitty 2520030137
Sahasra Reddy 2520030222



INTRODUCTION

Project Overview

The Nurse Shift Scheduling System is a healthcare management application designed to automate nurse scheduling across different wards and shifts.

The system ensures:

- Proper nurse allocation
- Skill-based assignments
- Leave management
- Overtime tracking
- Conflict detection
- Schedule notifications

Existing Problems

Manual shift scheduling is time-consuming.

Human errors in assigning nurses.

Difficulty in managing leave requests.

Uneven workload distribution.

Overtime calculations are complex.

Schedule conflicts may occur.

Proposed Solution

An automated scheduling system that efficiently manages nurse shifts while minimizing conflicts and improving workforce utilization.



Objectives of the Project

Automate nurse shift scheduling.

Match nurses based on skills and qualifications.

Manage leave requests efficiently.

Track overtime hours.

Detect scheduling conflicts automatically.

Generate reports and schedules.

Improve hospital workforce management.

System Architecture

Frontend (React.js)



REST API (Node.js + Express.js)



MySQL Database



Admin & Nurse Users

Components:

User Interface

Backend Server

Database

Notification System



Technologies Used

- Frontend
- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- Bootstrap / Tailwind CSS

Backend

- Node.js
- Express.js
- Database
- MySQL

Tools

- Postman
- npm
- Axios
- React Router
- Redux / Context API

Modules Implemented

1. Nurse Profile Management

Maintain nurse details and qualifications.

2. Roster Builder

Generate shift schedules automatically.

3. Shift Assignment

Assign nurses to wards and shifts.

4. Skill Matching

Allocate nurses based on required skills.

5. Leave Request Management

Handle leave applications and approvals.

Additional Modules

6. Overtime Tracker

Monitor overtime working hours.

7. Conflict Alert System

Detect overlapping shifts and scheduling issues.

8. Notifications

Send shift updates and alerts.

9. Export Schedule

Download schedules in PDF/Excel format.

10. Admin Panel

Manage nurses, schedules, and reports.

Database Design

- Tables
- Nurse
- Nurse_ID
- Name
- Qualification
- Skill
- Contact

Shift

- Shift_ID
- Shift_Name
- Start_Time
- End_Time
- Leave_Request
- Request_ID
- Nurse_ID
- Leave_Date

Status

- Schedule
 - Schedule_ID
 - Nurse_ID
 - Shift_ID
 - Date
-

Workflow

Admin logs in.

Adds nurse information.

Creates shifts.

Assigns nurses based on skills.

Nurses submit leave requests.

System checks conflicts.

Schedule is generated.

Notifications are sent.

Reports are exported.

Key Features

Automated scheduling

Skill-based nurse allocation

Leave management

Overtime calculation

Conflict detection

Real-time notifications

Report generation

User-friendly interface

Advantages

- Saves scheduling time.
- Reduces human errors.
- Improves staff utilization.
- Better shift management.
- Accurate overtime tracking.
- Enhanced patient care support.
- Easy report generation.

Future Enhancements

- AI-based schedule optimization.
- Mobile application support.
- Attendance integration.
- Biometric authentication.
- Cloud deployment.
- Analytics dashboard.

Conclusion

The Nurse Shift Scheduling System provides an efficient solution for managing nurse schedules, leave requests, overtime tracking, and conflict detection. It improves hospital operations by ensuring proper staffing and reducing administrative workload.

Thank You

